EXPERIMENT NO: 1

TITLE: Study of Power Semiconductor Devices and Their Applications in Industrial Electronics & Measure and analyze I-V characteristics using appropriate test circuits

AIM: To study various power semiconductor devices such as SCR, MOSFET and IGBT used in industrial electronics & To measure and analyze their I–V characteristics using suitable test circuits and understand their practical applications.

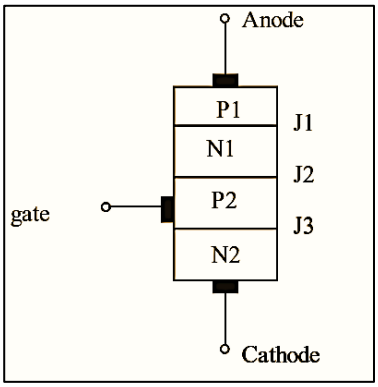
APPARATUS:

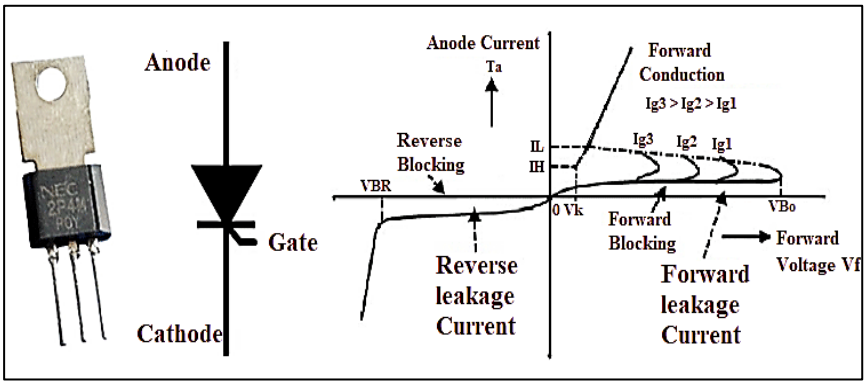
Sr.No.	Equipment	Specification	Quantity
1	Trainer Kit	Power Electronics Trainer by Micro Technologies	1
2	V-Lab	Virtual Lab (IIT-Delhi)	Online Software

THEORY: Power semiconductor devices are essential components in industrial electronics used for controlling and converting electrical power efficiently. They operate as switches or rectifiers in power electronic circuits, enabling control over large voltages and currents with minimal losses.

1. **Silicon Controlled Rectifier (SCR):**

   The SCR is a four-layer (PNPN) semiconductor device that acts as a controlled switch. It conducts current only when a gate pulse is applied and remains ON until the current falls below a certain holding value. SCRs are widely used in controlled rectifiers, AC voltage regulators, and motor speed control circuits due to their high current and voltage handling capability.





Working principle

- **Off-state (forward blocking):** When a positive anode-to-cathode voltage (V_{AK}) is applied with no gate current, the SCR blocks the flow of current, similar to an open switch. A small leakage current flows until the forward breakover voltage (V_{BO}) is reached.
- **On-state (forward conducting):** Conduction is triggered by a positive gate current pulse, which turns the SCR ON. It remains ON, or "latched," even if the gate signal is removed, as long as the current stays above the holding current (I_H).
- **Turn-off:** The SCR turns off when the anode current is reduced below the holding current level.

I–V characteristics measurement

Circuit diagram:
A basic circuit for testing SCR characteristics includes:

- A variable DC power supply (V_S) for the main anode-cathode circuit.
- A variable DC power supply (V_G) to provide the gate trigger pulse.
- A load resistor (R_L).
- A series ammeter (I_A) to measure anode current.
- A voltmeter (V_{AK}) in parallel with the SCR to measure anode-cathode voltage.
- A push-button switch in the gate circuit to apply a trigger pulse.

Procedure:

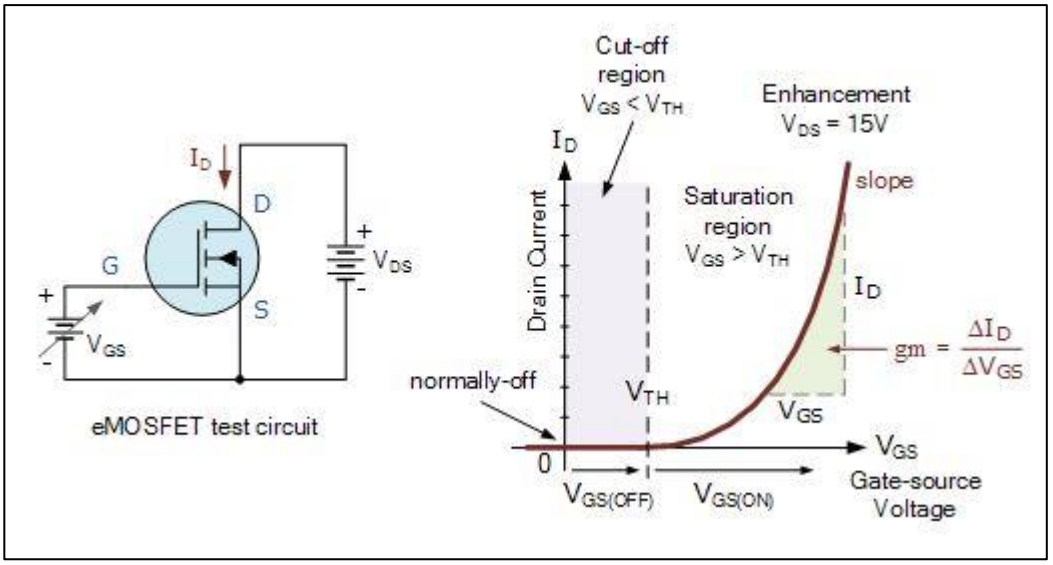
1. Connect the circuit with the gate switch open. Vary V_S and record the small leakage current while the SCR remains in the forward-blocking state.
2. Press the gate switch to trigger the SCR. Once the SCR turns ON, the anode current (I_A) increases sharply, and the voltage (V_{AK}) drops to a very low value.
3. Vary the gate current (I_G) and observe how it affects the forward breakover voltage (V_{BO}).
4. Remove the gate pulse and gradually decrease V_S to find the holding current (I_H), where the SCR turns off.

Practical applications

- **Motor speed control:** Used in both AC and DC motors for precise speed regulation.
- **Heating control:** Regulates power to electric heaters.
- **Power supplies:** Used in regulated power supplies and battery charging circuits.
- **Phase control:** Enables control over the percentage of AC current delivered to a load, as in a light dimmer.

2. Metal-Oxide-Semiconductor Field Effect Transistor (MOSFET):

A MOSFET is a voltage-controlled device used primarily for high-speed switching applications. It offers high input impedance and low switching losses, making it suitable for applications like DC–DC converters, inverters, and SMPS. Power MOSFETs are preferred in low to medium power applications where high frequency operation is required.



Working principle

- **Voltage-controlled operation:** A voltage applied to the gate terminal controls the current flow between the drain and source terminals.
- **Enhancement mode (most common):** Applying a positive gate-source voltage (V_{GS}) creates a conductive channel between the drain and source, turning the MOSFET ON. A larger V_{GS} widens the channel and increases current flow.
- **Off-state:** When V_{GS} is below the threshold voltage (V_T), the channel does not form, and the MOSFET acts as an open switch with high resistance.

I–V characteristics measurement

Circuit diagram:
A circuit for measuring MOSFET I-V characteristics includes:

- A variable DC power supply (V_{DD}) for the main drain-source circuit.
- A variable DC power supply (V_{GG}) for the gate-source circuit.
- A series ammeter (I_D) to measure drain current.
- A voltmeter (V_{DS}) in parallel with the MOSFET.

Procedure:

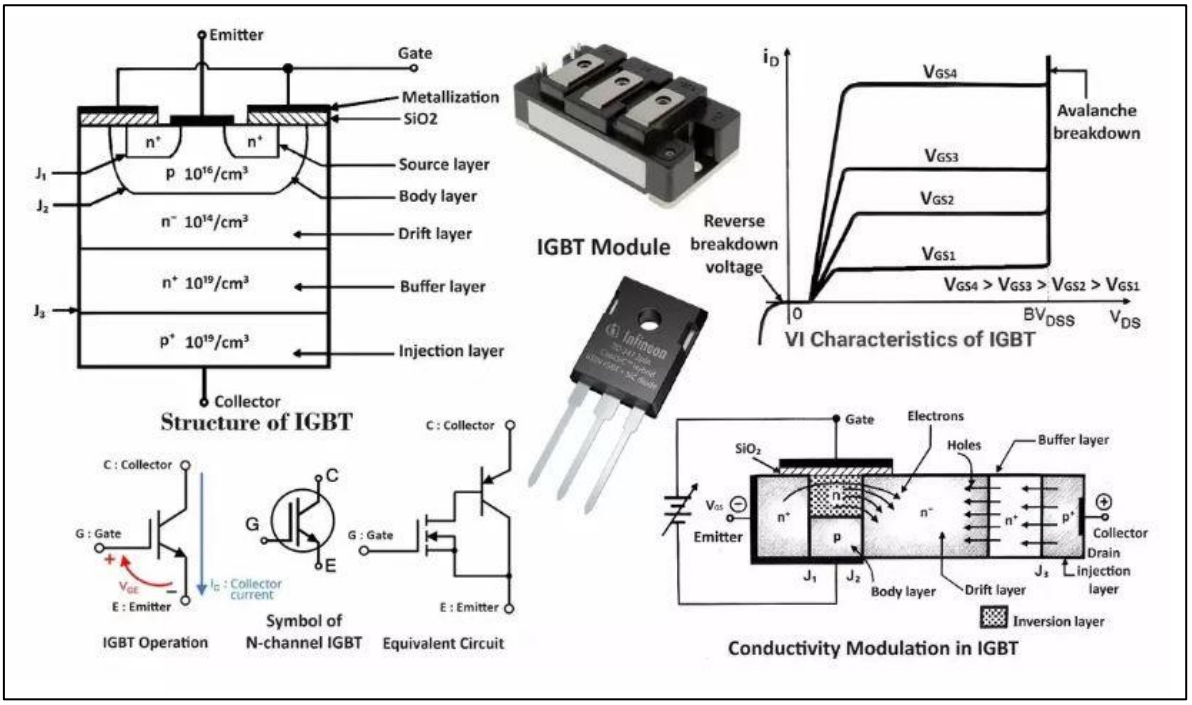
1. **Output characteristics (I_D vs. V_{DS}):**
    1. Set V_{GS} to a constant value, ensuring it is greater than the threshold voltage (V_T).
    2. Vary V_{DD} and record corresponding values of drain current (I_D) and drain-source voltage (V_{DS}).
    3. The plot will show a linear (ohmic) region at low V_{DS} and a saturation region where I_D becomes relatively constant.
    4. Repeat this process for several different constant values of V_{GS} to generate a family of curves.
2. **Transfer characteristics (I_D vs. V_{GS}):**
    1. Set V_{DS} to a constant value.
    2. Vary V_{GS} from 0 V upward and record the drain current (I_D).
    3. The plot will show that I_D remains zero until V_{GS} exceeds the threshold voltage (V_T)

Practical applications

- **Switch-mode power supplies (SMPS):** Their fast switching speed makes them ideal for power converters.
- **Automotive electronics:** Used in fuel injection, motor control, and power steering systems.
- **Audio amplifiers:** Utilized in high-frequency amplifiers.
- **Industrial drives:** Controls motors in various industrial equipment.

3. Insulated Gate Bipolar Transistor (IGBT):

The IGBT combines the advantages of a MOSFET's high input impedance and a BJT's low on-state conduction losses. It is a voltage-controlled device used in medium to high-power applications such as induction heating, traction drives, and inverters. IGBTs are preferred for high-efficiency power conversion at moderate switching frequencies.



Working principle

- **Combined operation:** Like a MOSFET, it is a voltage-controlled device that uses a gate voltage to control conduction. However, it also uses minority carriers (like a BJT) to achieve high current density and low ON-state voltage drop.
- **Conduction:** A positive gate-emitter voltage (V_{GE}) greater than the threshold voltage (V_{GET}) turns the device ON, allowing current to flow from collector to emitter.

I–V characteristics measurement

Circuit diagram:

A circuit for measuring IGBT characteristics includes:

- A variable DC power supply (V_{CC}) for the main collector-emitter circuit.
- A variable DC power supply (V_{GE}) for the gate-emitter control circuit.
- A series ammeter (I_C) to measure collector current.
- A voltmeter (V_{CE}) to measure collector-emitter voltage.

Procedure:

1. **Output characteristics (I_C vs. V_{CE}):**
    1. Set V_{GE} to a constant value.
    2. Vary V_{CC} and record the corresponding collector current (I_C) and collector-emitter voltage (V_{CE}).
    3. Repeat the process for several different constant values of V_{GE}.
    4. The curves will resemble those of a BJT, with an active region and a saturation region, controlled by the gate voltage.
2. **Transfer characteristics (I_C vs. V_{GE}):**
    1. Set V_{CE} to a constant value.
    2. Vary V_{GE} and record the collector current (I_C).
    3. The plot will show that no current flows until V_{GE} exceeds the threshold voltage (V_{GET}).

Practical applications

- **Motor drives:** Controls electric motor speed in applications like electric vehicles (EVs).
- **Renewable energy:** Used in inverters for solar and wind power systems.
- **High-voltage power transmission (HVDC):** Utilized in power grid equipment.
- **Induction cooking:** Found in induction cookers and other high-power consumer devices.

In this experiment, the I–V characteristics of SCR, MOSFET, and IGBT are studied using suitable test circuits to understand their conduction, switching behavior, and control mechanisms. The analysis of these characteristics helps in selecting appropriate devices for specific industrial applications.

RESULT:

Put the screenshot of the result of the experiment performed.(Each result page should contain Name, Roll No & Batch Name)

CONCLUSION:

Should be handwritten